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Marx's Hypotheses on the Length of the Working Day

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In order to argue that value depends exclusively on labor, Marx had to assume a uniform rate of surplus value (exploitation) across all industries. But if this rate of exploitation is the same everywhere, hours of work have also to be uniform. Marx accordingly insisted that this was so and quoted contemporary evidence in support. A close examination of his sources (nineteenth-century factory inspectors' reports) fails, however, to confirm his prediction. Instead the data are more consistent with the competing neoclassical hypothesis, originating with Jevons, that work hours will vary because workers' labor and leisure preferences vary.

I. Introduction

Of all the testable hypotheses that can be extracted from *Capital*, none has been more neglected than that concerning Marx's analysis of the length of the working day.¹ Many interpreters offer a simple verbal summary that typically runs as follows: Suppose a worker produces output to the value of his own wages in 6 hours. If the capitalist

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¹ Gotthilf (1966), who aimed at presenting a comprehensive collection of Marx's predictions, apparently overlooked the fact that his propositions on the working day translate into a whole set of clearly testable hypotheses. Eagly's (1974) contribution has been more helpful because his restatements of Marx's arguments in mathematical form lend themselves to empirical verification (although Eagly himself did not urge the need for it). Equations (1) and (2) borrow from Eagly's notation.

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imposes a working day of 10 hours he will be able to appropriate surplus value (measured in terms of labor) of 4 hours. If he lengthens the working day to 12 hours, while keeping wages constant, he will increase the exploited surplus labor from 4 to 6 hours, or by 50 percent. This is what Marx called *increasing absolute surplus value*.

Absolute surplus value obviously cannot grow without limit. Workers after all have a finite working capability. In any case the destruction of labor power through overwork is not consistent with maximum exploitation.

But the value of the labour-power includes the value of the commodities necessary for the reproduction of the worker, or for the keeping up of the working-class. If then the unnatural extension of the working-day, that capital necessarily strives after in its unmeasured passion for self-expansion, shortens the length of life of the individual labourer, and therefore the duration of his labour-power, the forces used up have to be replaced at a more rapid rate and the sum of the expenses for the reproduction of labour-power will be greater; just as in a machine the part of its value to be reproduced every day is greater the more rapidly the machine is worn out. It would seem therefore that the interest of capital itself points in the direction of a normal working-day. [Marx 1975, 1:266]

Since labor power in *Capital* is the exclusive source of potential surplus (and total) value, some reduction of an "excessive" working day, say 12 hours, in pursuit of some optimal length (from the capitalists' viewpoint) of say 10 hours, is therefore rational. But, according to Marx's arguments, such reduction usually calls for legislation to enforce it. His reasoning appears to have been as follows. Suppose one capitalist, operating independently, voluntarily reduces the working day to 10 hours. This isolated act will confer an advantage on the other (free-riding) employers who keep to the customary 12 hours. Since in the longer run it is unlikely that the exceptional employer who attempts 10 hours will be able to compete with his rivals, he will ultimately have to reinstate the 12-hour day. Only if all employers simultaneously are compelled by law to observe the shorter day will the reduction of hours succeed. This indeed is one of the inherent "contradictions" of capitalism.²

² The following passage is taken from Marx (1975, 1:270). "Capital that has such good reasons for denying the sufferings of the legions of workers that surround it, is in practice moved as much and as little by the sight of the coming degradation and final depopulation of the human race, as by the probable fall of the earth into the sun. In every stock-jobbing swindle every one knows that some time or other the crash must

If the legislated working day is successfully enforced, capitalists will concentrate on an alternative policy: cutting down the labor time necessary to produce the equivalent of the workers' wages. Suppose that exactly 6 hours are needed to create the amount of necessary value and the working day is legislated to have a maximum of, say, 10 hours. The induced adoption of better methods and machines may so improve labor productivity that only 4 hours of a worker's time need eventually to be devoted to producing his subsistence and the margin for surplus is increased. This is called increasing *relative* surplus value (Marx 1975, 1:509–10). In practice, of course, producers do not have to wait for legislated limitations on hours before they seek productivity improvements. They will normally pursue absolute and relative surplus value simultaneously. In any case, despite the legal limitations, entrepreneurs soon learn to cheat by adding minutes before and after the legal workday and by curbing the legally established time for meals (Marx 1975, 1:240–42).

The formulation of the working day is ultimately seen as the outcome of a conflict of "two competing rights": the rights of the capitalists to prolong the working day to its maximum length and the rights of workers to reduce it to one of a more normal duration. This contest between the two sides is the essence of the class struggle (Marx 1975, 1:270).

So much for the usual expositions. Further scrutiny reveals deeper issues. Marx (1975, 1:534) defines the rate of surplus value as:

$$e = \frac{\text{unpaid labor}}{\text{paid labor}}.$$

As Morishima (1973) observes, the rate of exploitation is equalized throughout the Marxian model of capitalism, and this follows only "if the length of the working-day is equalised throughout the economy, as is implicitly assumed in the definition of e above" (Morishima 1973, p. 52). If in fact the rates of exploitation between industries i and j differ because the hours in the working day, T_i in industry i ($= T_j$) are greater than T_j , there will be a migration of workers from i to j until an equilibrium is reached where $T_i = T_j$. In Marx's words: "This [the equalization of the rate of exploitation] would assume competition among labourers and equalisation through their continual migration from one sphere of production to another" (1975, 3:175).

There was an additional reason why Marx insisted on the same

come, but every one hopes that it may fall on the head of his neighbour, after he himself has caught the shower of gold and placed it in safety. *Après moi le déluge!* is the watchword of every capitalist and of every capitalist nation. Hence Capital is reckless of the health or length of life of the labourer, unless under compulsion from society."

duration of the working day for all workers. If individual employees could (and did) choose different hours of work, this would strike at the heart of his labor theory. It would imply that the standard unit of labor was a unit of disutility and not the objective "expenditure of human brains, nerves and muscles" that Marx described. In other words, he had to assume a uniform rate of surplus value across all industries and occupations in order to demonstrate that value depends on "objectified" labor exclusively. It is understandable, therefore, why Marx refers only obliquely in one or two places to Adam Smith's theory that a competitive labor market equalizes not the rate of wages for homogeneous labor but the "net advantages" of different jobs (Blang 1980).

Already we are in a position to test Marx's propositions empirically. Section II will first restate the Marxian hypotheses more precisely. Section III will then review the historical data available to Marx and quoted by him in support of his argument. A competing neoclassical hypothesis of labor markets and industrial organization will also be tested to see if it makes a more persuasive explanation of the same data. Section IV will explore some dynamic versions of Marx's theory and will focus on (a) the adduced positive relation between cyclical fluctuations and the length of the working day and (b) the positive relation between the length of working day and the accumulation of capital. Marx did not support these propositions with systematic contemporary evidence because this evidence was not available to him. Accordingly, they will be tested by applying appropriate twentieth-century data.

II. The Hypotheses

Marx himself might not have approved of the modern "scientific" tests of his various hypotheses as we are outlining them. Indeed, it has recently been argued that, unlike some of his followers, he was hesitant to apply the word "science" to his writings. Notice that he attacked the classical economists and the Utopian socialists mainly for not penetrating beyond the immediate facts. At the same time he denied the fact-value distinction. He brought to the facts a theory of consciousness and a vision of societies "as systems of relations among human beings, of which the relations entered into for purposes of 'production and reproduction' are primary" (Thomas 1976, p. 10).

This recent playing down of the scientific nature of Marx's work may have some validity, but it is focused on an attempt to reduce the central place in it of dialectical inevitability and "prediction" in the historical or prophetic sense of the word (Carver 1975, pp. 148–50). The fact is that important scientific aspirations do remain in Marx's

work. Certainly he made use of a vast amount of empirical knowledge, and it is from this emphasis that Marx seems to have considered his own work at least more scientific than that of the political economists (Carver 1975, p. 158). This being so, it is appropriate to observe that for any debate to be productive all participants must reach some basic agreement on the starting facts. Marx may have criticized the classical economists for not seeing through and beyond them, but the facts themselves constitute the common point of departure for all (Meek 1976). One of the tasks of this paper, therefore, is to examine how accurate Marx was in reporting them. A partial answer here, in the context of available nineteenth-century facts on "the working day," will be undertaken in due course.

Blaug (1968) argues that although Marx was much more aware of methodological issues than classical economists such as Ricardo, he made no conscious effort in his writings to verify his conclusions or to check his predictions against the available body of data. According to the same authority, Marx used the data in order to have a powerful effect on the reader and to suggest that the harsh nineteenth-century conditions depicted were a necessary product of capitalism, so that similar conditions will be found wherever capitalism prevails. Maintaining a proper historical perspective, we should be careful to ask, Blaug insists, what conclusions can be legitimately drawn from the evidence presented, especially in chapter 10 of *Capital* on "The Working Day." In particular, we should carefully inquire whether conditions in nineteenth-century Britain, as described in Marx's historical chapters, signal what he calls exploitation of labor or what others would diagnose as the low per capita output of a working population in a country still at a relatively early stage of its development. For, after all, "the living standards of the British working class could not have been raised significantly even by a perfectly egalitarian distribution of income" (Blaug 1968, p. 278).

The approach here is the same as Blaug's, although there will be an additional focus upon the accuracy of Marx's facts. In this and the next section I shall concentrate, first, on the particular documentary evidence that Marx used and, second, on a retrospective test of Marx's theories using a more systematic application of facts from his own data source. In other words, I will attempt to use the facts reported in *Capital* in the way Blaug intimates that Marx should have used them.

Before I come to Marx's evidence (examined in the next section) it will be helpful to summarize the basic structure of his model. The supply of hours of homogeneous labor (H^s) is equal to the number employed (N) and the number of hours in the working day (h). The supply schedule thus reads:

$$IF = hN. \quad (1)$$

The part of his work (in terms of labor hours) that constitutes the worker's daily subsistence (h^*) is called by Marx "necessary labor." Obviously, $h^* < h$. The worker's subsistence wage (in terms of hours of output) is the product of h and some ratio (k), so that $h^* = kh$, where

$$0 < k < 1. \quad (2)$$

The assumption of a fixed subsistence wage operates throughout Marx's discussion of the labor market. The assumption, of course, is merely an assertion that the wage payment is fixed "per unit of time" such as the long-run equilibrium fixed wage in traditional classical economics. For Marx, however, the elimination of the Malthusian equation and the emphasis on the newly introduced concept of an economy-wide unemployment rate permitted explicit variation in (a) the length of the working day and (b) the hourly wage. So, while the daily wage is fixed at subsistence, the hourly wage varies as the length of the working day varies. It has been argued accordingly (Eagly 1974) that in the wage bargain it was as if the employer saw his workers as motivated only by a specific subsistence requirement. In other words, it was as if the worker's supply of labor schedule had an elasticity of -1 , which is consistent with a constant (fixed) real wage.³

Marx's own words from the first volume of *Capital* illustrate the position: "From the law: 'the price of labour being given, the daily . . . wage depends on the quantity of labour expended,' it follows, first of all, that, the lower the price of labour, the greater must be the quantity of labour, or the longer must be the working-day for the labourer to secure even a miserable average wage. The lowness of the price of labour acts here as a stimulus to the extension of the labour-time" (Marx 1975, 1:548).

What bargaining there is over the length of the working day, Marx describes in terms of the class struggle:

We see then, that, apart from extremely elastic bounds, the nature of the exchange of commodities itself imposes no limit to the working-day, no limit to surplus-labour. The capitalist maintains his rights as a purchaser when he tries to make the working-day as long as possible, and to make, whenever possible, two working-days out of one. On the other hand, the peculiar nature of the commodity sold implies a limit to its consumption by the purchaser, and the

³ This paragraph has benefited from correspondence with Robert Eagly.

labourer maintains his right as seller when he wishes to reduce the working-day to one of definite normal duration. There is here, therefore, an antinomy, right against right, both equally bearing the seal of the law of exchanges. Between equal rights force decides. Hence is it that in the history of capitalist production, the determination of what is a working-day, presents itself as the result of a struggle, a struggle between collective capital, *i.e.*, the class of capitalists, and collective labour, *i.e.*, the working-class. [1:234–35]

It transpires that in the process of bargaining for the working day, workers have very few degrees of freedom. But whatever the bargaining strengths, the ultimate outcome is a working day of homogeneous duration. The final hypothesis in this section, therefore, is summarized in the proposition that the variance of hours will be almost negligible.

It is interesting to note that the neoclassicals were divided on the question whether the working day was inevitably of uniform length. When Böhm-Bawerk attacked Jevons's theory of disutility, he argued that workers cannot choose their hours, as Jevons implied, because they are fixed by capitalists (Stigler 1968). Jevons nevertheless insisted that *different* equilibrium hours of work would emerge as the outcome of differences in individual preferences (Jevons 1905). The equilibrium of utility and irksomeness is reached in each individual case when increments of utility and pain are equal.

Edgeworth (1925) supported Jevons's reasoning by observing that it applied to the real world. After all, flexibility of hours worked could be obtained through piece work, voluntary overtime, and choice between occupations. Marshall (1920) gave similar support to Jevons, in opposition to Böhm-Bawerk, by claiming that workers could seek from among different localities in order to select their individually desired hours. In the same tradition, but with more precision, Stigler (1966) argued that if some workers wanted, say, an 8-hour day and others wanted 10 hours, firms would emerge to accommodate these preferences and incorporate the appropriate machinery. The reason is that there is a trade-off between hours and wages for both employer and employee. A dispersion of hours and machine specifications can therefore be predicted.

It is clear that Marx took the same view as Böhm-Bawerk. Marx saw all the capitalists as maximizing all the hours of work and all the workers as minimizing all the hours of work, ignoring the potential trade-off between hours and wages that was available to both sides. The predictions of all participants in this dispute are clearly testable empirically. Again, we shall employ the same data source as that used in *Capital*.

III. Marx's Evidence

Marx argued that the greed of capital for surplus value was verified by the resistance of several manufacturers to the mid-nineteenth-century Factory Acts. These acts increasingly placed restrictions on the capitalist's power to extend the working day. Observing that "nothing is more characteristic of the spirit of capital than the history of the English Factory Act from 1833 to 1864," he leaned heavily for his illustrations on the reports of the British factory inspectors: "Certain guardians of these laws are appointed, Factory Inspectors, directly under the Home Secretary, whose reports are published half-yearly, by order of Parliament. They give regular and official statistics of the capitalistic greed for surplus-labour" (Marx 1975, 1:239).

Much of Marx's illustration consisted of selected inspector's reports of the behavior of individual mill owners. Such practice presents us with the difficulty of assessing the representativeness of anecdotal material. Part of the evidence, however, is much more systematic; and Marx himself seems to have recognized that this part required the most serious attention. It related to the work of Factory Inspector Leonard Horner who "conducted in his own person, and through his sub-inspectors, many examinations of witnesses in the factories of Lancashire" (Marx 1975, 1:285).

Marx assumed that nearly all adult male workers favored the imposition of statutory limitation on working hours, as had already been done for women and children. These desires were in conflict with those of the employers, who were continually attempting to maximize hours of work (and therefore surplus value). The workers' interest should be in reducing this degree of exploitation. Marx believed that Inspector Horner's survey supported this interpretation.

The survey was taken in October 1848 after two successive acts had reduced the maximum work limit for young persons (males 13–18 years old) and females, first to 11 hours, and then to 10 hours per day. Although the working day of adult male operatives was not regulated, the legislation had the effect, in some cases, of reducing their hours also. In Marx's words, Horner's survey showed that "about 70% of the workpeople examined declared in favour of 10 hours, a much smaller percentage in favour of 11, and an altogether insignificant minority for the old 12 hours" (Marx 1975, 1:285).

Horner's actual summary of the general results is reproduced here as table 1. An examination of them reveals three important errors in Marx's reporting of Horner's findings.¹ First, the proportion prefer-

¹ Another error (of much less significance) was that Marx argued that 10,270 adult male laborers in 181 factories were examined, whereas the correct figure was 1,153. Horner reported 10,270 in all the districts who were working more than 10 hours. His special survey, just described, came from a sample in the total area in his jurisdiction.

TABLE I
WORKERS' PREFERENCE FOR HOURS OF WORK, 1848

WORKERS INTERVIEWED	HOURS OF WORK PREFERRED		
	10	11	12
Men (<i>N</i> = 651)	441 (67.74)	96 (14.75)	114 (17.51)
Women (<i>N</i> = 502)	272 (54.18)	51 (10.16)	179 (35.66)
All (<i>N</i> = 1,153)	713 (61.84)	147 (12.75)	293 (25.41)

SOURCE.—Horner (1970); see App. of this article for original table.

NOTE.—Numbers in parentheses are percentages.

ring 10 hours' work was not 70 percent but 61.84 percent (see the first col. of the table). Second, the proportion preferring 12 hours' work was not "an altogether insignificant minority" but was a significant minority of 25.41 percent (see the third col.). Marx's third error was his failure to interpret Horner's verbal testimony clearly and comprehensively. In *Capital* it is asserted that, of the adult workers who were "made" to work overtime, the majority actually preferred working 10 hours for less wages but could not do so for fear of unemployment. This conclusion was based on Inspector Horner's observation:

It will be seen in the annexed evidence that the opinions of several men working twelve hours were taken and that *many* of them said that they would prefer working ten hours for less wages but they had no choice; that so many were out of employment, (so many spinners getting very low wages by having to work as piecers, being unable to do better), that if they refused to work the longer time, others would immediately get their places, so that it was a question with them of agreeing to work the long time, or of being thrown out of employment altogether. [Horner 1970, p. 17; emphasis added]

Whether this was the opinion of many (Horner's word) or of the majority (Marx's word) of the surveyed operatives working overtime is perhaps a moot point. What is interesting is that Horner's survey showed not only that there were operatives working 12 hours who preferred working 10 hours but also that there were others working 10 hours who preferred to work 12. The fact that such opposite

preferences existed makes ambiguous Marx's claim that there was universal coercion by employers.⁵

Data supplied by surveys, of course, suffer from well-known deficiencies. More reliable than stated preferences are actual choices made. In this sense, the most interesting statement in the whole of Horner's report, a statement that Marx did not quote, is the following:

It has however been again and again stated to me by mill-owners, that those who wish to work their factories more than ten hours a day have no difficulty in finding adult males to enable them to do so; that at all times, when a mill was working longer hours than its neighbours, it was always sure to draw to it the *best* hands; and that at this time most mill-owners who cannot, from the nature of their manufacture, work without young persons and women, and therefore not more than ten hours a day, are losing some of their *best* workers among the adult males, by their going to mills where they can get higher wages by working twelve hours. [Horner 1970, p. 16; emphasis added]

This statement clearly describes a process of search, mobility, and job change, and all at the primary initiative of workers. Such changes were not prompted by the threat of unemployment. Workers were leaving one employer for another of their own volition.

Despite the conventional deficiencies of surveys, and in fairness to Inspector Horner, it should be remarked that his was a particularly careful piece of work. A simple question to employees asking whether they would prefer 10 hours to 12 hours might have led some of them to assume that they would get the same total wages, in which case they would naturally choose 10 hours. Horner avoided such confusion by directing his questions thus: "You say that you are getting ten shillings a week for ten hours, and that you would get twelve shillings if you worked twelve hours; but although you only now get ten shillings you work two hours less, and so have two hours a day more at your own disposal. Now tell me, whether, taking all things into account, you would prefer to work ten hours for ten shillings a week, or go back to twelve hours to get more wages?" (Horner 1970, p. 14). Or, as an alternative, he asked, "Suppose there were three mills near together,

⁵ Marx's interpretation that the fear of unemployment was the main incentive to accepting long hours of work is not obvious. Horner's statement about so many who were "out of employment" is amplified by the words in his immediate parenthesis: "so many spinners getting very low wages by having to work as piecers, being unable to do better." This implies not unemployment but *alternative* employment, in adjacent skills but at lower wages.

in each of which the work was the same, and all equally convenient to you, one of them working ten hours a day where you would get ten shillings a week, another working eleven hours where you would get eleven shillings and the other working twelve hours where you would get twelve shillings, which of the three would you like to work in?" (p. 14).

Marx's hypothesis of near zero variance in hours calls for facts on actual hours rather than on worker preferences. Such evidence appears in Horner's report on the page following that quoted by Marx (on worker preferences). In Horner's words:

A very considerable number of mills in my district are or have been working more than 10 hours a day during the last half-year, some 11, some 12, some 13 hours and even more. I requested the Sub-Inspectors to make me a return of all the mills in their respective divisions, which, during the last half year, that is, since the 'Ten Hours' Act came into operation they found working more than 10 hours a day with adult males; and I find the aggregate number to be 181 mills and that the number of adult males so employed amounts to 10,270, the hours of work varying from 11 to 15 hours a day. [Horner 1970, pp. 16–17]

Horner did not report what proportion of the total were the 181 mills. In his report for the half year ending December 1841, however, he stated that he had 1,541 mills on his visiting books. If this figure was approximately the same in 1848 (although there is no precise evidence on this), about 12 percent of the mills would have been working at various hours more than 10. Horner (1842) also suggests that there were many mills working at various hours less than 10. The evidence available to Marx, therefore, did not support his hypothesis that capitalism was producing a fairly uniform working day; or, following Eagly (1974), the evidence refuted the Marxian implication that it was as if the supply of labor schedule universally had an elasticity of -1 .

It should be noticed, incidentally, that for Eagly's version of Marx's reasoning to hold, workers should have the same preference for the common subsistence wage and thereby for a common workday. The data in table 1, however, show that this did not occur in the real world. These data produce in fact a wide (95 percent) confidence interval estimate for the mean hours of work, assuming a normal distribution, of $8.906 \text{ hours} < u < 12.356 \text{ hours}$.⁶

⁶ In order to obtain a lower bound on the 95 percent confidence interval estimate of u , Chebyshev's theorem was applied. The resulting interval was $9.73 < u < 11.54$. So even when a normal distribution is not assumed the spread is still significant.

Consider next an extension of Stigler's (1966) model of multiple choices of hours of work. A plausible assumption is that in the early stages of industrial development there was a high income elasticity of demand for leisure. Facts supporting such an assumption should show that workers with higher weekly wages, such as older, more experienced, or more skilled workers, would be demanding shorter hours of work and that these demands were being accommodated.

Another part of Horner's survey bears out these predictions. Of the lowest paid men (receiving only 6–11 shillings per week), 50 percent were in favor of working more than 10 hours of work. More detailed evidence relating to a continuous functional relation between wages and hours of work can be found in table 2 of the appendix to Horner (1970).

Horner also revealed considerable variation in factory size and sophistication of machinery (see esp. Horner 1842). Such flexibility of machinery and hours of work again provides stronger corroboration for Stigler's hypothesis that if a majority of workers (in a given firm) prefer longer or shorter hours, employers will find it profitable to cater to this demand "because they can get better or cheaper workers by doing so" (Stigler 1968, p. 265). According to this logic it would have been more costly for manufacturers to have forced their workers to work longer than they preferred. Horner expressed the same sentiments. "But if we can suppose that, from any motive of self-interest or caprice, a mill-owner would adopt a plan of working so injurious to his people, I think there can be little doubt that he would not be able to continue it long, and he would soon be left by all who could obtain work in other mills where the masters were more considerate; or at all events, that his best and most respectable hands would not submit to be so treated; and would leave him" (p. 9).

Clearly, in Horner's view, the bargaining power between employer and employees was not the one-sided affair that Marx typically showed in *Capital*.

Marx's argument that the state was attempting to help oppressed workers by shortening their hours through factory legislation, at first sight, seems paradoxical. For he regarded the state as the repository of power of the ruling and exploiting bourgeois class. But then he suggested the public-good argument whereby legislation would reduce free riding upon the common pool of labor, to the benefit of *all* producers. More plausible, however, is the argument that a limited group or coalition of manufacturers used political power to curb their competitors who were outside the coalition. Marx himself quoted businessmen in the Lancashire textile industry who urged their operatives to agitate for the 9-hour working day "and promised contributions in money to this end" (Marx 1975, 1:270, n. 2). Marvel (1977) has demonstrated in fact that the legislation was designed to place a

differential burden on a subset of textile manufacturers. Its leading lobbyists were the larger operators, using steam power, who wanted to enjoy increased rents from reductions in production. These reductions were concentrated among their smaller, more labor-intensive competitors who used waterpower.

IV. The Dynamic Setting

It will be recalled that Marx saw the final determination of the length of the working day as the outcome of a huge collective bargaining process between worker and employer collectives (the class struggle). But because, in Marx's view of capitalism, events are always in a state of flux, the bargaining circumstances are constantly changing. The working day can therefore change in length from a given (uniform) level in one period to another at a subsequent period. According to Marx, worktime varies inversely with the bargaining strength of the workers. In the slump the increasing size of the reserve army of the unemployed reduces labor's bargaining power. In the boom the converse situation prevails. In short, labor's bargaining strength and, correspondingly, the length of the working day change countercyclically over the business cycle.

More precisely, Marx's argument was that although during recessions there are a smaller number of jobs and less output, on the (fewer) days of the week worked, those employed have to work longer hours per day. "Crises during which production is interrupted and the factories work 'short time,' *i.e.*, for only a part of the week, naturally do not affect the tendency to extend the working-day. The less business there is, the more profit has to be made on the business done. The less time spent in work, the more of that time has to be turned into surplus labour-time" (Marx 1975, 1:241).

This hypothesis, that the length of the working day rises in the slump and falls in the boom, is obviously testable with evidence that has been made available since Marx's time. The statistical indicator work of the National Bureau of Economic Research in fact refutes it. Indeed, for many years now, the reduction in the length of the average workweek has been ranked among the better advance-warning indicators of recession (see esp. Moore 1961, chaps. 3, 6, 7, 10, 15, and 16).

For completeness, we should mention another hypothesis that has been drawn from Marx's argument. It is that with the accumulation of capital the length of the working day will be increased (Gottheil 1966). One encouragement for this interpretation is no doubt Marx's observation that "it is the absolute interest of every capitalist to press a given quantity of labour out of a smaller, rather than a greater num-

ber of labourers, if the cost is about the same. . . . The more extended the scale of production, the stronger this motive. Its force increases with the accumulation of capital" (Marx 1975, 1:635). The process is strengthened by the forced growth of the industrial reserve army whose conspicuous plight serves to weaken employee resistance to long hours. "The over-work of the employed part of the working-class swells the ranks of the reserve, whilst conversely the greater pressure that the latter by its competition exerts on the former, forces these to submit to over-work and to subjugation under the dictates of capital" (1:636). Systematic facts that have become available since Marx's time easily refute this kind of prediction.⁷ Using evidence that was accessible to Marx, Hopkins (1982) argues that working hours were not drastically increased with the coming of the new proletariat and the new machinery of the Industrial Revolution.

V. Summary and Conclusions

Several hypotheses on the length of the working day can be drawn from Marx's *Capital*. The most crucial for his labor theory of value is the proposition of near zero variance in the hours of work. Given that in equilibrium the wage rate is the same everywhere, hours of work have also to be uniform in order to demonstrate the Marxian uniform rate of surplus value. And Marx had to assume a uniform rate of surplus value in order to show that it depends only on labor, an implication that has been recognized, or emphasized, in Marxian literature only recently (Blaug 1980, p. 10). The intention here has been to test the hypotheses empirically.

The same data source on which Marx depended (the nineteenth-century factory inspectors' reports) has thus refuted his central hypothesis of a uniform length of the working day. In addition, the reporting of contemporary facts (in this context) in *Capital* has been shown to be inaccurate. Marx's other hypotheses, concerning the effect of cycles and capital accumulation upon the working day, have also been empirically refuted.

Ironically, Böhm-Bawerk, the leading nineteenth-century critic of *Capital* and its labor theory of value, agreed with Marx that hours of work were rigid in the real world. He therefore precluded the line of investigation that has been pursued in this article, the results of which have at least as serious implications for Marx's labor theory as those

⁷ For a summarization of long-term trends in average weekly hours in the century since Marx wrote, see Zeisel (1958). The trend has been continuously and consistently downward.

connected directly with the famous "great contradiction" that Böhm-Bawerk attempted to demonstrate.

Meanwhile, and equally ironic, Marx's own source of evidence has been shown to provide more support to the competing neoclassical argument of choice and variability in work hours that was first offered by Jevons and was developed later by writers from Edgeworth to Stigler.

It has been observed (Meek 1976) that Marx produced the kind of heroic economic system that the model builder erects on the basis of a preliminary examination of the facts. In so doing, he eventually adopts a Schumpeterian "vision" of the economic process. The model builder then proceeds to construct particular analytical devices that are not arbitrary "but are dependent upon . . . the nature of the primary facts which they are used to explain . . ." (p. 114). Obviously, if the primary facts are wrong, the whole analytic superstructure is brought into question. Clearly the substance of our discussion has been far from trivial.

Appendix

Workers' Preference for Hours of Work in 1848^{*}

Of the 1,153 workers interviewed:

713, or 61.84 percent, prefer 10 hours' work;

147, or 12.75 percent, prefer 11 hours' work;

293, or 25.41 percent, prefer 12 hours' work.

Of 651 men,

441, or 67.74 percent, prefer 10 hours' work;

96, or 14.75 percent, prefer 11 hours' work;

114, or 17.51 percent, prefer 12 hours' work.

Of 502 women,

272, or 54.18 percent, prefer 10 hours' work.

51, or 10.16 percent, prefer 11 hours' work.

179, or 35.66 percent, prefer 12 hours' work.

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^{*} Original form of table 1 data, Horner (1970).

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